

2026 AI in Science Repository

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Preface

Science has always been a social and collective endeavor. From the Manhattan Project to the Human Genome Project to the modern AI race, many of the most consequential advances in human knowledge have come not from lone investigators but from coordinated groups working toward shared problems. In the Fall 2026, 2 faculty, 1 postdoc, and 14 PhD students leveraged these principles to collectively make sense of one of the most pressing issues in contemporary research:

How is AI changing the social nature of scientific inquiry, for better and for worse?

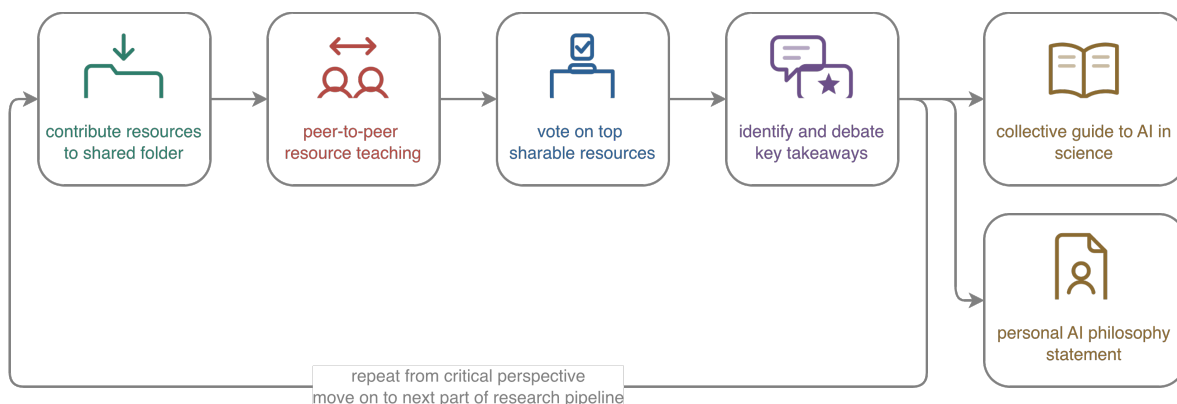


Figure 1: Structure of seminar on collective [artificial] intelligence in science.

The seminar began with an introduction to big-team science – its evolution, competitive advantages, and relevance to the current moment^{1–3}. From there, the course became a collective investigation, wherein we worked together to understand the newly disrupted nature of each stage of our research pipelines. This includes project management, literature review, idea development, data collection, analysis, writing, and dissemination⁴. In an attempt to develop a balanced perspective, we alternated between deploying receptive vs. critical lens, identifying key resources, consensus statements, and areas of disagreement along the way. Ultimately, the goal was to organize our insights into the resource you are viewing right now: the 2026 AI-in-Science Repository.

Each week followed a consistent structure (Figure 1). One component of our research pipeline would be selected for examination (e.g., data analysis). Members of the course would search for key resources for understanding the emerging role of AI. In class, we would then engage in

peer-to-peer teaching – all with the goal of identify the top 3 resources to include in the final repository. Last, we used a live online voting platform to solicit, debate, and vote upon key takeaways for the week (<https://pol.is/>). The result is the the 2026 AI-in-Science Repository: a summary of our top resources, takeaways, and disagreements.

1 Introduction

This is a book created from markdown and executable code.

See⁵ for additional discussion of literate programming.

Part I

Project Management

2 Project Management: Using AI

How is AI changing the way research teams plan, coordinate, and keep track of their work?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

2.1 Top Resources

The three resources the seminar voted to carry forward.

2.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

2.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

2.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

2.3 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

2.4 Open Questions

- *[question the week raised but could not answer]*

3 Project Management: Critiquing AI

What breaks when the coordination of a research team is delegated to AI?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

3.1 Top Resources

The three resources the seminar voted to carry forward.

3.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

3.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

3.3 Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

3.4 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

3.5 Open Questions

- *[question the week raised but could not answer]*

Part II

Literature Review

4 Literature Review: Using AI

How is AI changing the way we find, triage, and synthesize prior work?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

4.1 Top Resources

The three resources the seminar voted to carry forward.

4.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

4.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

4.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

4.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

4.3 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

4.4 Open Questions

- *[question the week raised but could not answer]*

5 Literature Review: Critiquing AI

What does AI-mediated search hide, distort, or invent — and how would we notice?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

5.1 Top Resources

The three resources the seminar voted to carry forward.

5.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

5.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

5.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

5.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

5.3 Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

5.4 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

5.5 Open Questions

- *[question the week raised but could not answer]*

Part III

Idea Development

6 Idea Development: Using AI

How is AI changing the way research questions and hypotheses are generated?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

6.1 Top Resources

The three resources the seminar voted to carry forward.

6.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

6.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

6.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

6.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

6.3 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

6.4 Open Questions

- *[question the week raised but could not answer]*

7 Idea Development: Critiquing AI

What happens to originality and intellectual ownership when ideas are co-generated?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

7.1 Top Resources

The three resources the seminar voted to carry forward.

7.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

7.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

7.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

7.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

7.3 Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

7.4 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

7.5 Open Questions

- *[question the week raised but could not answer]*

Part IV

Data Collection

8 Data Collection: Using AI

How is AI changing the way we recruit participants, build instruments, and gather data?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

8.1 Top Resources

The three resources the seminar voted to carry forward.

8.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

8.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

8.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

8.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

8.3 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

8.4 Open Questions

- *[question the week raised but could not answer]*

9 Data Collection: Critiquing AI

What threats to data integrity, validity, and research ethics does AI introduce?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

9.1 Top Resources

The three resources the seminar voted to carry forward.

9.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

9.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

9.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

9.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

9.3 Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

9.4 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

9.5 Open Questions

- *[question the week raised but could not answer]*

Part V

Analysis

10 Analysis: Using AI

How is AI changing the way we write code, model data, and interpret results?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

10.1 Top Resources

The three resources the seminar voted to carry forward.

10.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

10.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

10.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

10.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

10.3 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

10.4 Open Questions

- *[question the week raised but could not answer]*

11 Analysis: Critiquing AI

Which analytic errors survive when the analysis is delegated?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

11.1 Top Resources

The three resources the seminar voted to carry forward.

11.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

11.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

11.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

11.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

11.3 Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

11.4 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

11.5 Open Questions

- *[question the week raised but could not answer]*

Part VI

Writing

12 Writing: Using AI

How is AI changing the drafting, revising, and labor of scientific writing?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

12.1 Top Resources

The three resources the seminar voted to carry forward.

12.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

12.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

12.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

12.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

12.3 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

12.4 Open Questions

- *[question the week raised but could not answer]*

13 Writing: Critiquing AI

What is lost when writing stops being thinking — and who counts as an author?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

13.1 Top Resources

The three resources the seminar voted to carry forward.

13.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

13.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

13.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

13.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

13.3 Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

13.4 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

13.5 Open Questions

- *[question the week raised but could not answer]*

Part VII

Dissemination

14 Dissemination: Using AI

How is AI changing peer review, publishing, and how findings reach their audience?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

14.1 Top Resources

The three resources the seminar voted to carry forward.

14.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

14.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

14.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

14.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

14.3 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

14.4 Open Questions

- *[question the week raised but could not answer]*

15 Dissemination: Critiquing AI

What does AI do to trust, gatekeeping, and the integrity of the scientific record?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

15.1 Top Resources

The three resources the seminar voted to carry forward.

15.1.1 1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

15.1.2 2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

15.1.3 3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

15.2 Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

15.3 Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

15.4 Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

15.5 Open Questions

- *[question the week raised but could not answer]*

Part VIII

Personal AI Philosophy Statements

Personal AI Philosophy Statements

The collective guide in the preceding chapters records where the seminar converged. This part records where its members did not.

Alongside the shared repository, each participant wrote a personal AI philosophy statement: a short, signed account of how they intend to use — and refuse to use — AI in their own research. The statements were written independently, after the final week, and were deliberately not harmonized. Read together, they are a snapshot of how a single room of scientists, working from the same set of resources, arrived at meaningfully different positions.

The prompt

Participants were asked to address four questions in roughly 500-1000 words:

1. **Where do you delegate?** At which stages of your research pipeline do you currently use AI, and what do you gain by doing so?
2. **Where do you refuse?** What will you not delegate, and on what grounds — epistemic, ethical, professional, or personal?
3. **What would change your mind?** Name the evidence or capability that would move one of your refusals into your delegations.
4. **What do you owe others?** How will you disclose your use of AI to collaborators, reviewers, readers, and students?

Reading these statements

Each statement speaks only for its author. Disagreement between statements is the point, not an error to be corrected.

Template

Copy this structure into a new file under `philosophies/` and add it to the chapter list in `_quarto.yml`.

```
# [Your name] {.unnumbered}

*[Role - e.g., PhD student, Department of Psychology, University of Florida]*

## Where I delegate

## Where I refuse

## What would change my mind

## What I owe others
```

Nicholas A. Coles

[Role, department, institution]

Where I delegate

[Which stages of your pipeline, and what you gain.]

Where I refuse

[What you will not delegate, and on what grounds.]

What would change my mind

[The evidence or capability that would move a refusal into a delegation.]

What I owe others

[Disclosure to collaborators, reviewers, readers, and students.]

16 Summary

In summary, this book has no content whatsoever.

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